BMEN 600
Midterm Research Plan

Transcriptomic transferability between human brain regions

A machine-learning study of regional predictive sharing with a planned Parkinson's external test

Revised: October 12, 2026
Midterm deadline: October 16, 2026
Final project deadline: December 4, 2026

Team members: [INSERT ALL 3-4 TEAM MEMBER NAMES]
Working GitHub repository: [INSERT WORKING REPOSITORY URL]

Draft status. The scientific sections include verified preliminary analyses of the supplied GTEx files. Before submission, replace the identity and repository fields, confirm the proposed role assignments, and complete the actual author-contribution and team-verification statements. A functioning repository is a required component of the assignment.

Six pages of main text, excluding this title page and references.

1. Introduction and Motivation

Parkinson's disease, Alzheimer's disease and amyotrophic lateral sclerosis (ALS) affect particular neuronal populations and brain regions unevenly. This selective vulnerability raises a biomedical question: why do some neural systems degenerate while others remain comparatively preserved? Molecular and cellular properties help frame this problem, but regional vulnerability cannot be reduced to one expression pattern [1], [2]. Better characterization of regional molecular organization could support research into disease mechanisms and the design of more informative experimental models.

Obtaining matched transcriptomic measurements from multiple regions of a human brain is difficult. Consequently, it is useful to determine how much information measured in one region can predict expression in another. Brain atlases demonstrate anatomical organization of gene expression [3], [4], while GTEx supplies molecular measurements across tissues from the same donors [5]. A predictive analysis can therefore test whether source-region expression captures variation in target-region expression among people whose data were not used to train the model.

Research question: To what extent can machine learning quantify transcriptomic transferability between human brain regions, and how are differences in transferability associated with molecular characteristics relevant to regional vulnerability in neurological disease?

Transferability is defined as held-out-donor prediction performance. The main analysis compares directional region pairs and molecular characteristics. A planned Parkinson's extension will test a separate GTEx-trained frontal-cortex gene-prediction model on independent disease-stage data [16]. Because that dataset samples one region, it cannot validate cross-region predictions. The contribution is a reproducible assessment of regional predictive relationships and a clearly bounded external test of gene-expression prediction.

The biomedical engineering relevance is assessing whether transcriptomic measurements support useful computational proxies and whether those relationships generalize to independently collected disease tissue. Researchers could use the results to prioritize follow-up experiments. Patients and caregivers motivate the disease context, but neither regional prediction nor an external Parkinson's test establishes a diagnostic biomarker or a mechanism of vulnerability. Clinical interpretation requires independent biological validation.

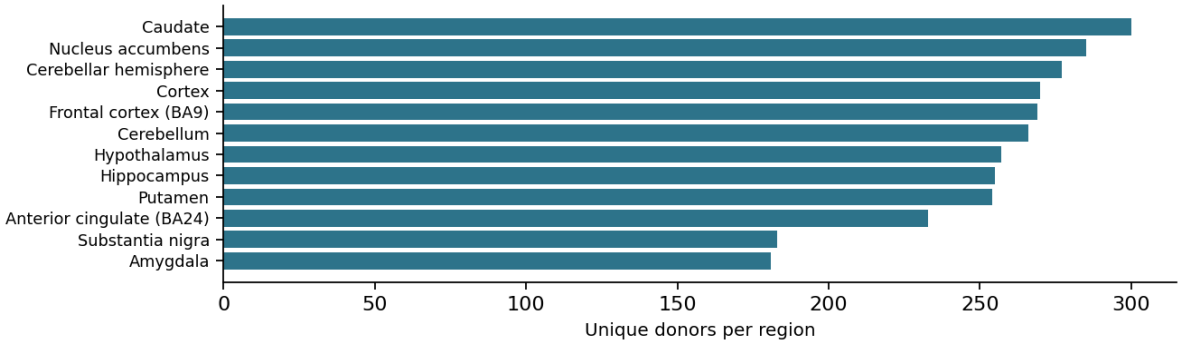


Figure 1. Available unique donors in the 12 brain-region files, calculated from the supplied data. Unequal coverage motivates pair-specific donor matching and cohort-size reporting. The separately uploaded spinal-cord file is excluded from the main analysis.

2. Background and Research Gap

Selective vulnerability provides the biological motivation, but it also defines the limits of expression-based inference. Fu, Hardy and Duff synthesize evidence that neuronal susceptibility reflects multiple intrinsic properties and disease-related processes [1]. In a more focused study, Fu *et al.* combined single-nucleus transcriptomic analyses with experimental investigations of tau accumulation, identifying a tau-homeostasis signature and implicating BAG3-related proteostasis [2]. Together, these studies motivate molecular comparisons while showing that cell-specific and experimental evidence is needed to connect expression with degeneration. Bulk regional measurements alone cannot establish the same mechanisms.

Spatial transcriptomic resources demonstrate that the human brain has structured molecular organization. Hawrylycz *et al.* used anatomically detailed microarray sampling to build an adult human brain atlas [3]. Their work enables comparisons across finely defined anatomical locations, although deep sampling in a small number of donors limits population-level inference. Subsequent analysis identified reproducible regional expression signatures associated with cellular and anatomical features [4]. These findings support looking for shared and region-specific molecular patterns; reproducibility of anatomical signatures, however, does not directly quantify whether one donor's regional expression profile can predict another region in a new donor.

GTEx complements spatial atlases by sampling multiple tissues across many individuals. The consortium used bulk RNA-seq and genetic association analyses to map shared and tissue-specific regulatory effects [5]. Its matched-donor design makes cross-region modelling feasible, but regional missingness and heterogeneous tissue composition complicate interpretation. Kim-Hellmuth *et al.* combined inferred cell-type enrichment with genotype to characterize cell-type interaction effects on expression [6]. Their results show why apparently shared or tissue-specific regulatory signals can change when cellular composition is considered. The present study therefore treats cell mixture as a plausible alternative explanation for predictive patterns, rather than interpreting every performance difference as a distinct neuronal program.

The closest computational precedent is BRONTE, which integrates GTEx and the Allen Brain Atlas using learned gene representations and brain-ontology information to impute expression in additional tissues [7]. It evaluates held-out expression measurements and the recovery of biological patterns. This demonstrates the promise of incorporating anatomical information into expression prediction, but its multi-region inputs and ontology-based model differ from a directional, single-source-region comparison. The version cited here is a preprint, and its results will be interpreted with that status explicit. This project does not reproduce BRONTE or perform its downstream genetic-risk analyses.

The manageable gap is an interpretable comparison of directional regional prediction under consistent evaluation, followed by a bounded test of whether GTEx-trained gene relationships remain predictive in independent disease tissue. Training-only selection and unseen-donor evaluation will support this comparison. The project does not claim that cross-tissue prediction is new or that lower prediction accuracy proves selective vulnerability.

The small disease panels provide context rather than vulnerability labels [8]-[10]. Cappelletti *et al.* examined postmortem frontal-cortex RNA-seq across Lewy-body pathology stages, adjusting for demographic, RNA-quality and cellular-composition effects [16]. Expression changes were most pronounced at Braak Lewy-body stage 5, motivating stage-aware evaluation rather than pooling all pathology groups. This observational, single-region study supports an external gene-prediction extension but supplies no measured second region for cross-region validation.

3. Dataset and Responsible Use

The training resource is Adult GTEx V11, supplied as regional gene-read-count matrices and inspected on October 9, 2026. GTEx Portal documentation describes the V11 GENCODE 47 annotation update [11]; the consortium paper describes V8 background [5]. The team will record the original download date and exact export provenance. The planned external resource is GSE216281 from Cappelletti *et al.* [16]. Its published cohort comprises 84 frontal-cortex donors: 23 controls, 19 at Braak Lewy-body stages 1-4, 19 at stage 5 and 23 at stage 6. It includes incidental Lewy-body disease; pathology-positive donors should not all be labelled clinical PD. These are published counts, not a verified downloaded cohort.

The 13 supplied GCT files each contain 74,628 gene rows. Header and sample-ID checks identified 3,234 tissue records from 393 donors. Excluding cervical spinal cord leaves 12 brain-region files, 3,030 records and 385 donors (Table 1). Within each region, every sample represents a distinct donor; donors recur across regions. Predictive outcomes are normalized expression and test error. One independent case is a donor, not a gene or tissue record.

Table 1. Sample coverage in the supplied matrices. Sample and unique-donor counts are identical within each region. Totals across regions count repeated donors more than once.

Region	Samples / donors	Main analysis
Amygdala	181	Included
Anterior cingulate (BA24)	233	Included
Caudate	300	Included
Cerebellar hemisphere	277	Included
Cerebellum	266	Included
Cortex	270	Included
Frontal cortex (BA9)	269	Included
Hippocampus	255	Included
Hypothalamus	257	Included
Nucleus accumbens	285	Included
Putamen	254	Included
Spinal cord (cervical C-1)	204	Excluded
Substantia nigra	183	Included

For GTEx, eligibility requires both regions in a donor and at least 60 paired donors per direction. For the external dataset, sample IDs, diagnosis, pathology stage, region and available covariates must be audited against the paper before inclusion. Tissue was sampled from frontal cortex (BA9 or BA10), limiting correspondence with GTEx BA9 [16]. The external independent case is one donor's frontal-cortex sample; there is no measured paired target region. Final sample counts and exclusions remain pending data retrieval.

GTEx separates public processed expression from protected data [12]. We will not attempt re-identification or use protected genotypes. Public materials will contain code and aggregate outputs; original matrices and donor-level files will remain excluded while applicable sharing conditions are checked. The external study reports informed consent and ethics approval [16]; its article licence does not by itself establish every dataset-sharing permission.

Both resources are postmortem bulk tissue. Sampling, age, sex, ancestry, cell mixture and RNA quality may affect prediction. GTEx covariates are absent from the attached matrices; the external paper describes a European-ancestry cohort [16]. Diagnosis/pathology and laboratory effects must not be conflated. Permitted metadata will be sought and residual confounding acknowledged.

4. Proposed Methods and Evaluation

The current Colab notebook loads GTEx GCT counts, validates dimensions and finite nonnegative values, checks library totals and matches donor IDs. Counts are transformed to log2(CPM + 1); this exploratory normalization is not TPM. Composition-aware normalization sensitivity will be assessed [13]. For each directional pair, 80% of matched donors form a development set and 20% remain untouched for testing (seed 42). The development set is split into 75% inner training and 25% validation; all records for a donor stay together within a model.

Inner-training data select 500 variable source genes and 1,000 targets after requiring CPM above 1 in at least 20% of donors and nonzero variance. Eligible disease-panel targets are added [8]-[10]. Training-only scaling and multi-output ridge regression are used [14]. Validation MSE selects a penalty from 1, 10, 100 and 1,000; the fixed feature lists are retained while the model is refitted on development donors. Test data never tune preprocessing or the model [15].

The primary regional result is median gene-wise test R^2: 1 - SSE/SST, with SST calculated around each gene's test-cohort mean. Negative values are retained and constant genes are undefined. A development-mean predictor supplies a practical baseline; skill = 1 - MSE(model)/MSE(baseline). Conditional 95% intervals use 1,000 bootstrap resamples of test donors with fixed predictions, not resampling genes. Repeated donor splits, common-gene/cohort comparisons and metadata sensitivity will precede definitive pair rankings.

Molecular analyses relate gene-wise predictability to development-set expression level, variance and regional mean-expression difference using descriptive Spearman associations. Disease-panel coverage is reported. These correlated-gene associations do not establish causality or pathway enrichment. The pilot covers six directions; the final regional analysis can extend to 132 eligible ordered pairs.

External Parkinson's test. GSE216281 supplies frontal-cortex measurements only [16]. A cross-region GTEx model cannot be scored without measured source and target regions from the same external donor. Accordingly, a separate model will predict a fixed set of frontal-cortex target genes from a disjoint set of measured frontal-cortex predictor genes. It will be trained and tuned on GTEx BA9, evaluated first on held-out GTEx donors, then frozen before any external scoring. A target gene must never appear among predictors. This tests cross-cohort gene prediction, not direct regional transferability or PD diagnosis.

Before fitting this secondary model, annotation-based Ensembl mapping will define a shared gene universe and resolve version suffixes and duplicates. The paper used GENCODE v35 and Salmon/tximport, unlike the GTEx export [16]. Count type, library preparation and scale will be audited. Both cohorts will be projected onto a fixed common gene universe and normalized consistently; scaling, gene selection and penalties are learned only from GTEx. Missing predictors will be reported rather than replaced with zeros. If compatible counts and complete predictors cannot be obtained, the external test will be reported as infeasible rather than scored misleadingly.

External controls will be scored first, then pathology groups separately. Median gene-wise R^2, RMSE and skill against the frozen GTEx mean baseline will be reported, with donor-bootstrap intervals and case/control error contrasts. Poor control performance signals technical/anatomical domain shift; poorer pathology-group performance alone cannot isolate disease effects. BA9 versus BA10, age, sex, RNA quality and cell composition will be assessed when metadata permit. Stage-specific comparisons are exploratory and do not tune the model.

Python, pandas, NumPy, scikit-learn and Matplotlib implement the current GTEx pipeline. Planned external loading, gene mapping and evaluation modules will reuse its exports and record sample inclusion, configuration and package versions. External evaluation has not yet been implemented or executed; GTEx pilot findings are retained as preliminary evidence.

5. Preliminary Results

A local execution of the AI-assisted notebook on the supplied files completed six directional models. The paired cohorts contained 194-233 donors, with 39-47 donors held out per direction (Table 2). The matrices passed the notebook's dimension, nonnegative/finite-value and nonzero-library checks. Table 1 and Figure 1 establish adequate matched regional coverage for a pilot while showing that sample availability is unequal.

Table 2. Verified pilot performance. R^2 is the median of gene-wise held-out-donor scores. Skill uses the development-mean baseline. Each direction evaluates approximately 1,000 training-selected targets plus eligible disease-panel genes.

Prediction direction	Paired / test	Median R^2	Skill
Caudate to Putamen	233 / 47	0.450	0.457
Putamen to Caudate	233 / 47	0.385	0.347
Cortex to Frontal cortex (BA9)	194 / 39	0.496	0.472
Frontal cortex (BA9) to Cortex	194 / 39	0.363	0.386
Cerebellum to Cerebellar hemisphere	222 / 45	0.294	0.379
Cerebellar hemisphere to Cerebellum	222 / 45	0.272	0.346

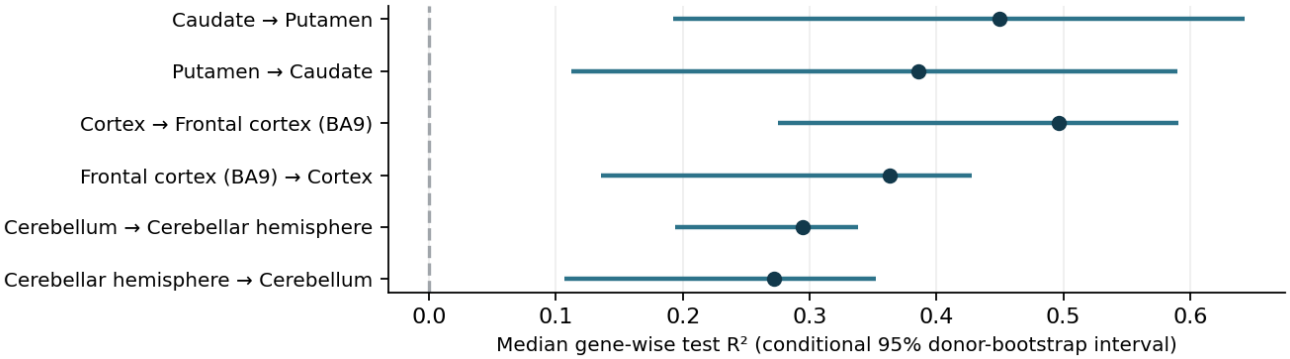


Figure 2. Median gene-wise test R^2 and conditional 95% intervals from 1,000 donor-bootstrap resamples of fixed predictions. Overlapping intervals, different cohorts and different targets preclude a definitive regional ranking.

Median gene-wise R^2 ranged from 0.272 to 0.496; skill ranged from 0.346 to 0.472. The pilot therefore demonstrates that the pipeline can recover target-region variation beyond the simple mean baseline for these selected genes and splits. The direction-dependent scores suggest that transferability is not symmetric under this model, but they do not establish intrinsic biological asymmetry because target composition and variance also differ.

For caudate-to-putamen, regional mean-expression difference had a descriptive Spearman association of -0.376 with gene-wise R^2; the reverse direction was -0.408. The association was weaker or absent in other pilot directions, including cortex-to-frontal cortex (0.013). All 13 prespecified disease genes were evaluated in each pilot direction. These observations motivate broader and controlled comparisons, rather than a universal conclusion about regional differences or vulnerability.

These pilot results support feasibility of the regional analysis, but no GSE216281 data have been downloaded or scored in this project. External performance cannot be inferred from the GTEx results. Remaining work is to audit the external cohort, align genes and scales, implement and freeze the separate frontal-cortex predictor, and evaluate controls before pathology groups. The main uncertainty is biological versus technical domain shift. This cohort cannot provide direct cross-region PD validation; that would require another dataset with donor-matched regions.

6. Teamwork and Project Plan

The proposed allocation below separates biological interpretation, data stewardship and modelling while requiring joint review. Role labels must be replaced with team-member names before submission. For a three-person group, the analysis lead also coordinates reproducibility; for four members, that role can be assigned separately. These assignments are proposed responsibilities, not claims of completed contributions.

Table 3. Proposed milestones and owners through the final submission.

Dates	Milestone / evidence	Proposed owner
Oct. 9-16	Review citations and pilot outputs; complete midterm; publish working notebook/README and confirm access.	Literature lead; data lead; analysis lead
Oct. 17-30	Retrieve/audit GSE216281; record V11 provenance; align gene IDs, counts and permitted covariates.	Data lead; analysis lead
Oct. 31-Nov. 13	Run regional sensitivity tests; train/freeze the separate GTEx frontal gene predictor; score external controls and pathology groups.	Analysis lead; reproducibility lead
Nov. 14-20	Check stage-specific external errors and molecular associations; assess batch, anatomy and cell-mixture explanations.	Literature lead; analysis lead
Nov. 21-27	Draft paper, captions and limitations; independently rerun central output and review repository instructions.	All members
Nov. 28-Dec. 4	Revise paper; rehearse presentation; verify links; freeze graded repository commit and submit.	All members

Data and Code Availability

GTEx source: gtexportal.org/home/datasets/. Planned external dataset: GEO GSE216281 [16]. Authors' code: chiaracapp/Transcriptomic-profiling-of-PD-brains. Our working repository: **[INSERT WORKING GITHUB URL]**. The existing notebook produces the GTEx pilot; the external model remains planned. The repository must provide a README, setup/access instructions, current code and organized aggregate results. Original matrices and donor-level PRIVATE outputs remain excluded pending sharing review.

AI-Assisted Work

ChatGPT assisted with literature retrieval, notebook development, debugging, preliminary data analysis, graph generation and drafting this proposal. The analysis code was executed on the supplied files, and the plotted pilot results were checked against the resulting tables during this session. This does not substitute for team verification. Before submission, team members must inspect cited sources, review donor-level splitting and training-only preprocessing, rerun at least a central output, and check that the written claims match it. **Team verification completed:** [INSERT ACTUAL CHECKS, MEMBERS AND DATE].

Author Contributions

[Member A]: [Record actual research-question, literature and writing contributions]. **[Member B]:** [Record actual data-preparation, responsible-use and analysis contributions]. **[Member C]:** [Record actual modelling, code, figures and evaluation contributions]. **[Member D, if applicable]:** [Record actual reproducibility, interpretation and revision contributions]. All authors should confirm the final text and their recorded roles. Do not submit proposed responsibilities as completed work.

References

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